

Motion in vector form in two dimensions — Problem Set

Warm-up

1. Given the position vector $\mathbf{r}(t) = 3t\mathbf{i} + (4t^2 - 2)\mathbf{j}$, find the velocity vector $\mathbf{v}(t)$ at $t = 2$.
2. A particle moves with constant velocity $\mathbf{v} = 5\mathbf{i} - 12\mathbf{j}$. Calculate the speed of the particle.
3. Eliminate the parameter t to find the Cartesian equation of the path defined by $x = 2t$ and $y = t^2 + 1$.
4. Convert the displacement vector $\mathbf{s} = 3\mathbf{i} + 3\mathbf{j}$ into polar form (magnitude and direction as an angle θ from the positive x -axis).
5. If a boat heads due North at 4 m/s in a river flowing due East at 3 m/s, write the resultant velocity vector in component form.

Skill-building

1. A particle's position is given by $\mathbf{r}(t) = (\cos 2t)\mathbf{i} + (\sin 2t)\mathbf{j}$. Show that the particle moves in a circular path and find the radius of this path.
2. The velocity of a projectile is $\mathbf{v}(t) = 20\mathbf{i} + (50 - 10t)\mathbf{j}$. Find the time t when the projectile reaches its maximum height (i.e., when the vertical component of velocity is zero).
3. Find the Cartesian equation for a particle with parametric equations $x = e^t$ and $y = e^{2t} - 3$, stating any restrictions on x .
4. Two particles have position vectors $\mathbf{r}_1 = t\mathbf{i} + (2t + 1)\mathbf{j}$ and $\mathbf{r}_2 = (2t - 3)\mathbf{i} + (t + 2)\mathbf{j}$. Find the time t at which the particles collide, or show that they do not collide.
5. A swimmer aims to swim directly across a river of width 80 m. The swimmer's speed in still water is 2 m/s and the river current is 1.2 m/s. Calculate the time taken to reach the opposite bank if the swimmer maintains a heading perpendicular to the current.
6. A particle has velocity $\mathbf{v} = \frac{1}{t+1}\mathbf{i} + 2t\mathbf{j}$. If the particle starts at the origin when $t = 0$, find its position vector $\mathbf{r}$ at $t = 3$.
7. Show that the parametric equations $x = 3 + 2\cos\theta$ and $y = -1 + 2\sin\theta$ represent a circle. State the center and radius.
8. An object moves such that $\mathbf{r} = (t^3 - 3t)\mathbf{i} + (t^2)\mathbf{j}$. Find the coordinates of the point(s) where the particle is momentarily at rest in the x -direction.
9. A plane flies with an airspeed of 250 km/h on a bearing of 060° . A wind is blowing from the South at 50 km/h. Express the resultant ground velocity as a vector in $\mathbf{i}, \mathbf{j}$ form.
10. For the curve $x = \frac{1}{t}$ and $y = \sqrt{t^2 + 1}$, find the gradient of the path $\frac{dy}{dx}$ in terms of t .

11. A particle moves along the path $y = x^2 - 4x$. If the horizontal velocity component is constant at $\dot{x} = 3$ m/s, find the velocity vector when $x = 5$.
12. Prove that the distance d of a particle from the origin, where $\mathbf{r} = (5 \cos t)\mathbf{i} + (5 \sin t)\mathbf{j}$, is constant for all t .
13. Two objects P and Q have positions $\mathbf{r}_P = (t+1)\mathbf{i} + t^2\mathbf{j}$ and $\mathbf{r}_Q = (3-t)\mathbf{i} + (4t-4)\mathbf{j}$. Determine if their paths intersect and if they are at the intersection point at the same time.
14. A point P moves such that its position vector is $\mathbf{r} = (t-2)^2\mathbf{i} + (t-2)\mathbf{j}$. Sketch the path for $0 \leq t \leq 4$.
15. Find the acceleration vector $\mathbf{a}(t)$ for a particle whose position is $\mathbf{r} = (t - \sin t)\mathbf{i} + (1 - \cos t)\mathbf{j}$. Calculate the magnitude of the acceleration.

Easier Exam Questions

1. (Hurlstone 2024 Q11c)

(i) Eliminate t from the parametric equations:

1

$$\begin{aligned}x &= \cos t \\y &= \sin^2 t\end{aligned}$$

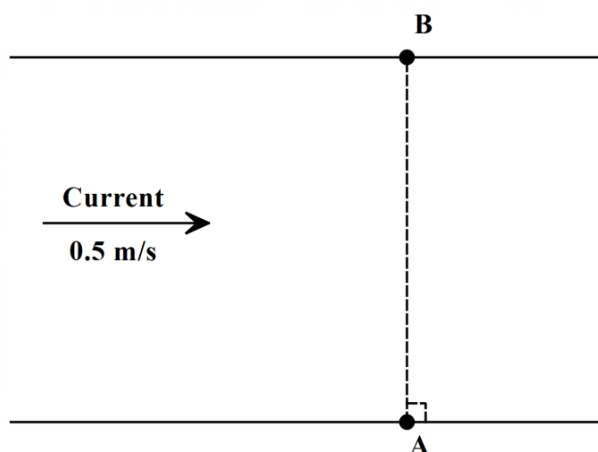
(ii) Sketch the curve defined by the parametric equations in (i)

1

2. (Newington 2024 Q13b)

In still water, Anthony can swim at 1.5 m/s. Anthony is at point A on the edge of a riverbank and considers point B directly opposite. A current is flowing from the left at a constant speed of 0.5 m/s.

(i) If Anthony dives in straight towards B, and swims without allowing for the current, what



will his actual speed and direction be?

2

(ii) Anthony wants to swim directly across the canal to point B. At what angle should Anthony aim to swim in order that the current will correct his direction?

2

3. (Hornsby Girls 2023 Q12e) The position of a boat, relative to a stationary observer at the origin, point O , is being recorded.

The boat travels at a constant speed and in a constant direction and passes through three collinear points A , B and C . The boat is first observed at point A on a position vector of $-20\mathbf{i} - 12\mathbf{j}$.

One minute later, the boat is observed at point B on a position vector of $-5\mathbf{i} + 5\mathbf{j}$.

Using binoculars and without taking their eyes off the boat, the observer watches the motion of the boat for five minutes, starting when the boat is at point A , passing through the point B and continuing through to the point C .

- (i) Find the position vector of the boat after the five minutes when it is at point C . 2
(ii) Hence, find $\angle AOC$, the angle through which the observer's binoculars move during this five minute interval, correct to the nearest degree. 2

4. (Hurlstone 2021 Q11d) Two boats on a lake start sailing at the same time.

Boat A moves on a course given by: $x = \frac{t}{2}$; $y = t + 1$.

Boat B moves on a course given by: $x = t - 2$: $y = 9 - 2t$.

where t measures the time elapsed in hours from when the boats started.

- (i) Find the Cartesian equation for the course of each boat and show that the courses intersect at $(1, 3)$. 2
(ii) Do the boats collide? Justify your answer supported by calculations. 1
5. (Killara 2020 Q13c) Sienna intends to row her boat from the south bank of a river to meet with her friends on the north bank. The river is 100 metres wide. Sienna's rowing speed is 5 metres per second when the water is still. The river is flowing east at a rate of 4 metres per second. Sienna's boat is also being impacted by a wind blowing from the south-west, which pushed the boat at 8 metres per second. She starts rowing across the river by steering the boat such that it is perpendicular to the south bank.

- (i) Show that the velocity of Sienna's boat can be expressed as the component vector:

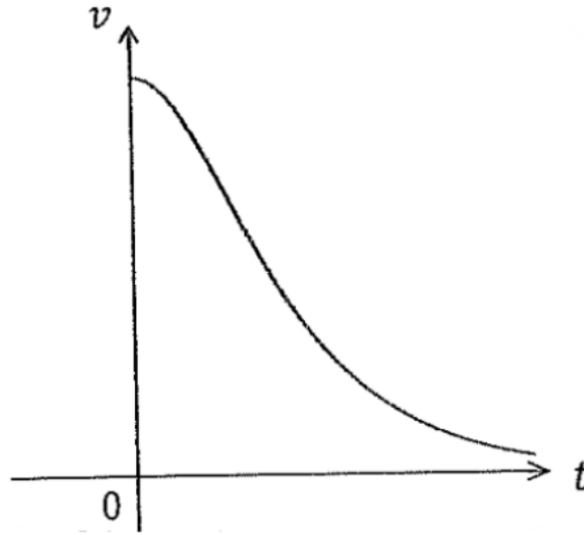
$$(4 + 4\sqrt{2})\mathbf{i} + (5 + 4\sqrt{2})\mathbf{j}$$

- 2
- (ii) Calculate the speed of the boat, correct to 2 decimal places. 1
(iii) Determine the distance rowed from Sienna's starting point to her landing point and how long it will take her to reach the north bank. 3

Harder Exam Questions

1. (Fort St 2024 Q14a) A particle P is moving along the x axis. At time $t = 0$, it was at $x = 0$. Its velocity $v \text{ ms}^{-1}$ at a time t is given by $v = \frac{e^t}{1+(e^t)^2}$. The graph below shows its velocity as a function of time. 4

Given that the displacement of the particle when $t = k$ seconds is $\tan^{-1} \frac{1}{2}$ metres, show that $k = \ln 3$.



2. (Ascham 2024 Q14b) Sketch the function represented by the parametric equations 3

$$x = 2t - 1 \quad y = t^2 + t \quad -1 \leq t \leq 1.$$

3. (Barker 2020 Q13d) A ball rolling along a horizontal plane has position and velocity vectors given below: Position in metres: $\underline{r} = x\underline{i} + y\underline{j}$, $t \geq 0$, $0 < y < 1$ Velocity in ms^{-1} : $\underline{\dot{r}} = 2y\sqrt{y}\underline{i} + y(1-y)\underline{j}$ The component of the velocity in the $\underline{j}$ direction is the differential equation $\frac{dy}{dt} = y(1-y)$. It can be shown that $\frac{1}{y(1-y)} = \frac{1}{y} + \frac{1}{1-y}$. Do not prove this.

- (i) Solve the differential equation $\frac{dy}{dt} = y(1-y)$ to find the general solution, and hence show that if the particle is initially at the point $(0, 0.5)$, the component of the displacement in the $\underline{j}$ direction is: 3

$$y = \frac{e^t}{1 + e^t}$$

- (ii) Show that the magnitude of the velocity vector is $y(1+y)$ and hence calculate the magnitude of the velocity at time $t = \ln 9$ seconds. 3

4. (Girraween 2020 Q15c) An equation can be expressed in the parametric form

$$x = 2\cos\theta - 1$$

$$y = 2 + 2\sin\theta$$

Express the equation in Cartesian form. 2

5. (Sydney Grammar 2020 Q11f) A particle moves in two dimensional space where $\underline{i}$ and $\underline{j}$ are unit vectors in the x and y directions respectively. At time t seconds its displacement from the origin is given by $\underline{r} = (6t - 4t^2)\underline{i} + 2t\underline{j}$ where all lengths are measured in metres.

- (i) Write down the particle's velocity vector in component form. 1

- (ii) Find the speed of the particle when $t = 2$. 1

- (iii) Show that the equation of the path of the particle is $x = 3y - y^2$. 2

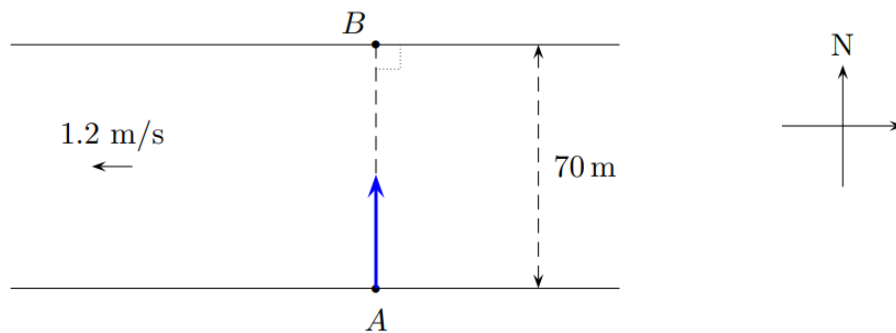
6. (Normanhurst Boys 2021 Q12c) The position of jet ski relative to a stationary ship is plotted on the ship's radar. One unit on the radar represents a distance of 1 metre along the sea level, and the origin represents the ship's position.

The jet ski travels at a constant speed and is observed at point A on a position vector of $-20\mathbf{i} + 36\mathbf{j}$. One second later, the jet ski is observed at point B on a position vector of $-2\mathbf{i} + 51\mathbf{j}$.

A crew member on the ship observes the motion of the jet ski using binoculars for ten seconds starting from point A , without taking his eyes off of the jet ski.

- (i) Find the position vector of the jet ski after the ten seconds. **2**
- (ii) Hence find the angle through which the crew member's binoculars moves through during this ten second interval, correct to the nearest minute. **2**

7. (Normanhurst Boys 2023 Q12e) A fish at point A wants to swim across the river that is 70 m wide. The banks of the river are parallel and the points A and B are on opposite sides of the river. **3**



The fish swims at a speed of 1.8 m/s and the river is flowing downstream at 1.2 m/s. However, a 0.5 m/s wind blows constantly to the north-westerly direction.

What is the bearing that the fish should head to in order to land directly at point B from point A ?